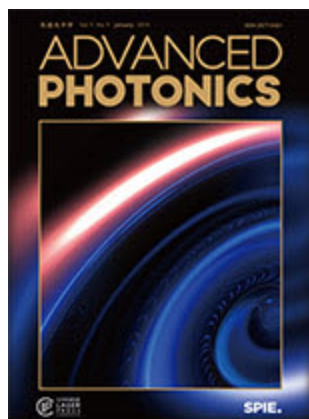




Supercontinuum Fourier-transform Overtone Photothermal Spectroscopy and Microscopy

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1 Supercontinuum Fourier-transform Overtone Photothermal 2 Spectroscopy and Microscopy

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8 **Abstract:** Overtone photothermal (OPT) microscopy is a promising modality for vibrational volumetric imaging in
9 aqueous samples, owing to reduced water absorption and increased penetration depth. However, the wavelength-
10 scanned, frequency-domain implementations impose a trade-off among spectral bandwidth, acquisition speed, and
11 spectral resolution, limiting rapid and information-rich chemical fingerprinting. Here, we introduce Fourier-transform
12 overtone photothermal (FT-OPT) spectroscopy and microscopy, a time-domain framework that retrieves photothermal
13 spectra without sequential wavelength tuning. In FT-OPT, interferometric modulation of a supercontinuum pump
14 excites broadband overtone transitions, and the resulting local thermal lens simultaneously encodes the vibrations into
15 a temporal interferogram. This signal is sensitively read out by a shorter-wavelength probe beam, thereby preserving
16 higher spatial resolution. Fourier transformation of the interferogram enables spectra spanning 3000 cm⁻¹, with
17 spectral resolution approaching the intrinsic molecular linewidths, millimolar sensitivity, and 2-3 orders-of-magnitude
18 faster spectral acquisition than wavelength-scanned OPT. Using FT-OPT, we discriminate molecules with closely
19 related overtone signatures, demonstrate time-resolved overtone fingerprinting in a microfluidic platform, and perform
20 label-free 4D (3D spatial and spectral) imaging of thick tissues. By overcoming the bandwidth-speed trade-off while
21 retaining high spatial resolution, FT-OPT provides a route toward label-free spectral flow cytometry and high-
22 dimensional chemical imaging in aqueous environments.

23 **Keywords:** Supercontinuum, Fourier-transform spectroscopy, Overtone vibration, Photothermal spectroscopy,
24 Photothermal microscopy

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1 Introduction

Vibrational spectroscopy and imaging^{1–4}, revealing intrinsic signatures of molecular structure, are indispensable tools for label-free and chemically specific detection across environmental science, biology, medicine, chemistry, and materials science. In particular, owing to its ultrahigh sensitivity, enhanced spatial resolution and optical sectioning capability, mid-infrared (MIR) photothermal microscopy⁵ has attracted increasing attention over the past decade^{6–10}. By using a shortwave beam to detect the local thermal lens effect¹¹ induced by MIR absorption under high numerical aperture (NA) conditions, this approach achieves micromolar sensitivity and submicron spatial resolution. Nevertheless, MIR-based methods require tunable and broadband sources such as quantum cascade lasers (QCLs), optical parametric oscillators (OPOs), or synchrotron radiation¹², as well as other specialized optical materials and components, which substantially increase system complexity and cost. More critically, the strong absorption of water leads to severe signal background and attenuation, often necessitating measurements under dry or dehydrated conditions^{13–15}. This effect constrains applications in aqueous environments and restricts penetration depth in complex biological tissues⁵. Recently, researchers proposed overtone photothermal (OPT) microscopy^{16,17} based on shortwave infrared (SWIR, 1000–2000 nm). While preserving the advantages of photothermal microscopy, exciting overtone transitions with a SWIR source dramatically suppresses water absorption¹⁷, which not only reduces system cost but also enables 3D imaging of biological samples at millimeter depths¹⁶.

Although demonstrating outstanding imaging ability, photothermal detection cannot obtain spectra via a normal spectrometer; instead, spectra are acquired by tuning the pump wavelength. This frequency-domain excitation approach introduces a series of constraints in spectral performance due to the limitations of mainstream tunable SWIR sources^{3,18}. On the one hand, the tuning range is difficult to cover several hundred nanometers or more¹⁹. As a result, existing OPT microscopy typically covers only a portion of the C–H overtone region¹⁷. Due to the notable spectral overlap of overtone and combination transitions, a broader spectral range is required to extract more chemically specific information²⁰. On the other hand, prevailing SWIR sources (e.g., OPOs) also suffer from limited tuning speeds: manual adjustment typically requires a few seconds¹⁷, and even most automated tunable sources operate on timescales of tens to hundreds of milliseconds per wavelength step. Owing to the positive coupling between acquisition time and spectral bandwidth, it is necessary to spend several seconds or minutes to acquire a spectrum, which is unsuitable for

dynamic spectral applications. Additionally, because most of these sources operate as ultrafast pulses¹⁹, their short durations fundamentally lead to spectral broadening due to the time–energy uncertainty principle¹⁷. While the broad linewidths in overtone regions reduce the requirement for spectral resolution, high spectral resolution remains advantageous for distinguishing similar molecules.

Inspired by FTIR and dual-comb spectroscopy²¹, time-domain excitation schemes have recently been explored in photothermal microscopy^{12,22} as a tentative approach to overcome the above trade-offs, but most remain at the proof-of-concept stage. Even in strong fundamental MIR absorption bands, their photothermal signals remain weak, hindering label-free detection. Therefore, existing demonstrations require thermosensitive fluorescent labeling and long integration (~10 min per spectrum). This performance not only cannot accommodate dynamic spectral monitoring or hyperspectral imaging, but also preserve the above limitations of MIR-based approaches. This may critically underestimate the potential of time-domain excitation in photothermal spectroscopy and microscopy.

To address these challenges, in this work, we present Fourier-transform overtone photothermal (FT-OPT) spectroscopy and microscopy, which replaces the conventional frequency-domain excitation with a time-domain interference strategy. To realize this concept, we employed a femtosecond laser and photonic crystal fiber (PCF) to generate a high-power supercontinuum as SWIR pump, enabling intense photothermal signals without the need for labeling or long integration times. By encoding and Fourier-transform decoding the local thermal lens dynamic, FT-OPT decouples the trade-off between spectral bandwidth and acquisition time, obtaining spectra with ultimate spectral resolution within hundreds of milliseconds. The parallel and photothermal measurement²³ also endows it with high sensitivity. The supercontinuum bandwidth encompasses the C–H 2nd and O–H 1st overtone regions, enabling clear discrimination and unmixing of C8 hydrocarbons, even in overtone regions with severe spectral overlap. We further develop the first microfluidic overtone spectroscopy for dynamic spectral monitoring. Moreover, we demonstrate hyperspectral (4D) photothermal imaging for the first time within a time-domain excitation framework.

2 Results

2.1 Principle of FT-OPT spectroscopy and microscopy

Molecular anharmonicity causes deviations from an ideal harmonic potential, giving rise to overtone and combination transitions that carry chemical specificity^{20,24}, as illustrated in Fig. 1(a). When a wavelength-matched SWIR pump is focused onto the sample by objective, these transitions are excited and subsequently relax non-radiatively, generating local heat. The resulting temperature rise produces a refractive-index gradient¹¹. Under the small-temperature-rise approximation, the refractive-index change Δn near the focal volume can be written as Eq. (1).

$$\Delta n \propto \int k(\omega) |E_{\text{pump}}(\omega)|^2 d\omega \quad (1)$$

Here, ω is the pump frequency, $k(\omega)$ is the molecular response coefficient, which directly corresponds to the overtone intensity spectrum, and $|E_{\text{pump}}(\omega)|^2$ represents the pump spectrum (spectral power density). Then, the thermal lens induced by Δn is detected by a separate shortwave probe that is collinear with the pump but slightly defocused. Conventional photothermal microscopy employs a single-wavelength pump³ and acquires spectra $k(\omega)$ by tuning the pump wavelength. In FT-OPT, two identical broadband SWIR pulses with a controlled relative delay τ are used as the pump-pair. Under this scheme, the refractive-index change $\Delta n(\tau)$ can be expressed by Eq. (2).

$$\begin{aligned} \Delta n(\tau) &\propto \int k(\omega) |E_{\text{pump}}(\omega) + E_{\text{pump}}(\omega)e^{i\omega\tau}|^2 d\omega \\ &= C_k + F^{-1}\{k(\omega) |E_{\text{pump}}(\omega)|^2\} + F^{-1}\{k(-\omega) |E_{\text{pump}}(-\omega)|^2\} \end{aligned} \quad (2)$$

Here, $F^{-1}\{\}$ denotes the inverse Fourier transform, $C_k = 2 \int k(\omega) |E_{\text{pump}}(\omega)|^2 d\omega$ is a constant.

By scanning the relative delay τ , the thermal lens effect is probed and $\Delta n(\tau)$ is recorded, forming an interferogram $I(\tau) \propto I_{\text{probe}} \Delta n(\tau)$. For a specific vibrational transition frequency, Figs. 1(b) and 1(c) illustrate two representative cases of τ . In Fig. 1(b), τ_c corresponds to constructive interference, yielding a pump-pair spectrum distribution that maximally excites the transitions, and the thermal lens response reaches a local maximum. In Fig. 1(c), τ_d corresponds to destructive interference, and the signal reaches a local minimum. The interval between every adjacent τ_c and τ_d corresponds to half a vibrational period. Ultimately, $I(\tau)$ concurrently encodes all vibrational states that can be excited within the pump spectrum, with each state modulating the photothermal signal at its

vibrational frequency, and the resulting FT-OPT interferogram $I(\tau)$ is shown by the red curve in Fig. 1(e). A subsequent Fourier transform $F\{\}$ of $I(\tau)$ then yields Eq. (3).

$$F\{I(\tau)\} \propto I_{probe} [C_k \delta(0) + k(\omega) |E_{pump}(\omega)|^2 + k(-\omega) |E_{pump}(-\omega)|^2] \quad (3)$$

The second term in Eq. (3) corresponds to the spectral information as shown by the blue curve in Fig. 1(e), which can be converted into the spectrum $k(\omega)$ by normalizing with the pump spectrum $|E_{pump}(\omega)|^2$. The reasoning presented above is general, and holds for both SWIR overtone and MIR fundamental transitions¹². Therefore, the spectral bandwidth of FT-OPT is determined by the pump spectrum, while the acquisition speed is governed by the scanning speed of the relative delay τ . Additionally, as long as the scanning range of τ is sufficient, FT-OPT can record the molecular vibrational lifetimes^{25,26}, and in principle, obtain the spectral features near the natural linewidth limit.

A major advantage of photothermal imaging is that it operates as a pump–probe “two-photon” process, inherently providing axial optical sectioning capability^{5,17}. Moreover, the lateral point spread function (PSF) is jointly determined by the pump and probe beams, allowing the use of a shorter-wavelength probe to further enhance spatial resolution. 4D hyperspectral images can be obtained by performing point-by-point scanning in the X–Y–Z volume based on the above pump–probe processes. Another advantage is that, particularly in overtone applications, photothermal microscopy offers a significant sensitivity advantage over Beer–Lambert–based methods. This is because overtone transitions are several orders of magnitude weaker than fundamental transitions, and the light–matter interaction length is extremely short (several micrometers), making the pump intensity modulation depth inherently limited. Photothermal detection improves sensitivity by converting weak absorption into a measurable thermal response, while time-domain excitation introduces a multiplex (Fellgett) advantage²³ that further enhances the signal. Taking lipids as an example, the present FT-OPT system achieves ~86-fold enhancement in signal modulation depth (0.075% vs 6.5%). A detailed analysis is provided in the Supplementary Information and Fig. S1.

2.2 SWIR supercontinuum generation and experimental setup

The FT-OPT experimental setup is schematically shown in Fig. 1(d), with detailed specifications and component models provided in the Methods section. Specifically, the output of a commercial fiber femtosecond laser was coupled into a segment of PCF. The intensity autocorrelation of the

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3 141 injected pulse and the PCF coupling setup is shown in Fig. S2. Due to the extremely high peak
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5 142 power of the femtosecond pulses and the strong nonlinear effects in the PCF, the pulses broadened
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7 143 spectrally^{27–29}, and the duration was stretched to the picosecond range by PCF dispersion to reduce
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9 144 optical damage to the sample. This process produced a high-power (~21 mW at the sample after
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11 145 50% duty cycle modulation), spectrally flat source spanning 1100–1600 nm (6250–9091 cm⁻¹),
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13 146 covering the entire C–H 2nd and O–H 1st overtone regions, where most organic molecules exhibit
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15 147 pronounced transitions and rich chemical specificity^{20,24}. The flat pump spectrum also ensures
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17 148 uniform and efficient excitation of all vibrational states within this range. The first-order
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19 149 autocorrelation of the pump, as well as the pump spectrum and the PCF input spectrum, are shown
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21 150 in Figs. 2(a) and 2(b), respectively. The broadband power calculation procedure, as well as the
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23 151 objective transmission and detector response curves, are provided in the Supplementary
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25 152 Information (Fig. S3).

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27 153 As shown in Fig. 1(d), the pump was then directed into a Michelson interferometer (MI) for beam
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29 154 splitting and relative delay scanning. A 1 mW continuous-wave He–Ne laser (632.8 nm) was split
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31 155 1:1 to serve as the probe beam and the relative delay reference beam, respectively. For detection,
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33 156 two identical objectives (NA = 0.65) were used to focus and collect the beams. An avalanche
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35 157 photodiode (APD) was used to measure the photothermal signal $I(\tau)$, while two photodetectors
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37 158 (PDs) monitored the relative delay τ and the first-order autocorrelation of the pump (with
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41 160 For signal acquisition, FT-OPT employs a classical photothermal microscopy approach in which
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45 162 (LIA); the overall signal processing workflow of the system is shown in Fig. 1(e). The pump was
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47 163 modulated by a chopper at $f_m=10kHz$ with a 50% duty cycle, and the relative delay τ was
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49 164 continuously and bidirectionally scanned at 2000 fs/s by a motorized scanning stage. Additionally,
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51 165 XY–Z scanning was performed using motorized translation stages at the sample and objective to
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53 166 enable volumetric imaging. The raw time-domain signals and the relative delay calibration
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60 167 procedure are provided in the Supplementary Information (Fig. S4).

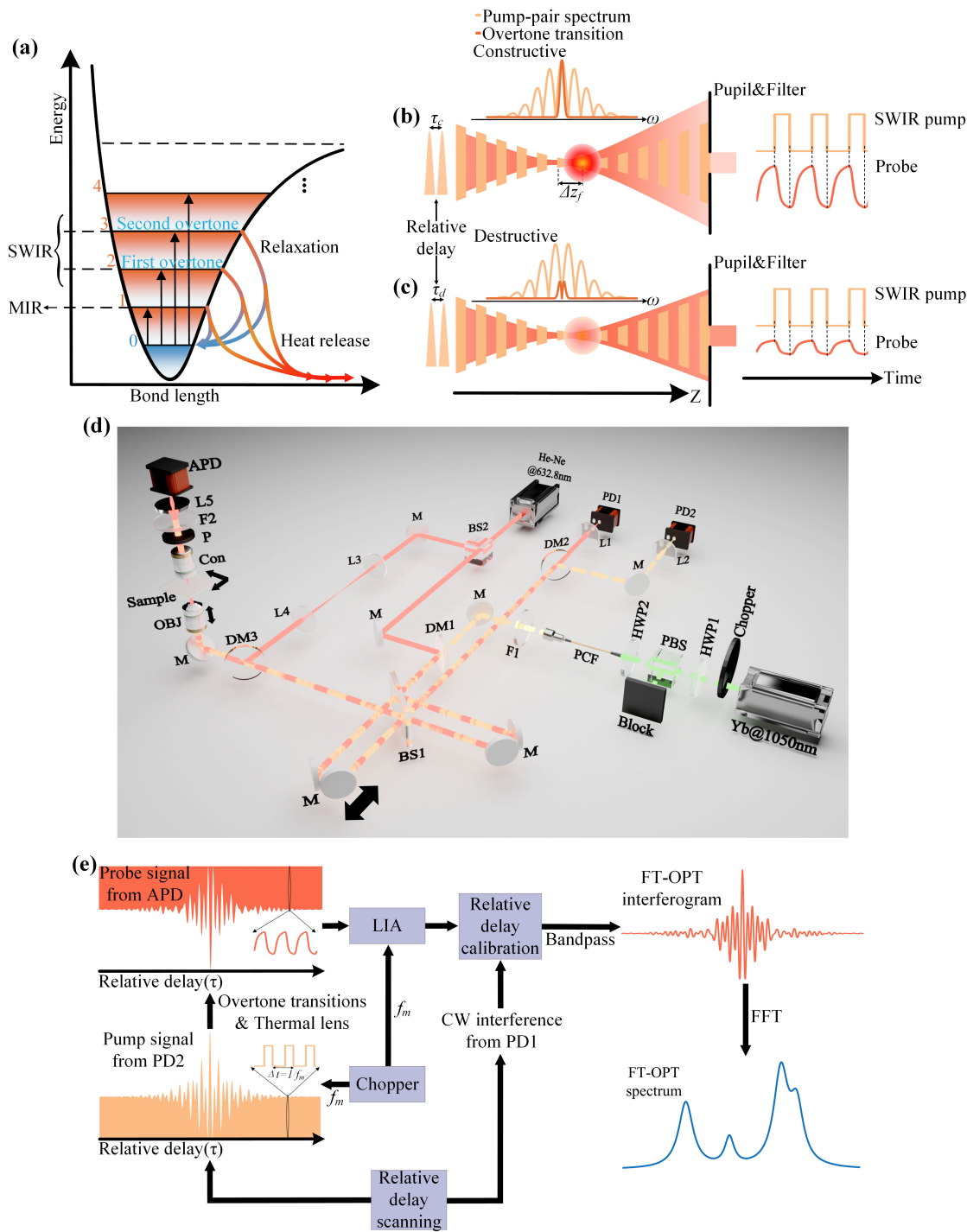


Fig. 1 Principle and setup of FT-OPT. (a) Schematic of SWIR overtone transitions, nonradiative relaxation, and heat release, where 0 and 1 denote the ground and fundamental vibrational states, respectively. (b, c) Pump-probe processes of FT-OPT. Two representative relative delays of the pump-pair for a specific vibrational state are shown, illustrating the thermal lens response with a local maximum or minimum in time-domain. (d) Experimental setup of FT-OPT. HWP: half-wave plate; PBS: polarizing beam splitter; PCF: photonic crystal fiber; F: filter; M: mirror; BS: beam splitter; DM: dichroic mirror; L: lens; OBJ: objective; Con: condenser; P: pupil; PD: photodetector; APD: avalanche photodiode. (e) Signal processing workflow of FT-OPT. CW: continuous-wave; f_m : pump modulation frequency.

2.3 System characteristics

FT-OPT was first performed on pure ethanol and 1-octyne samples to evaluate the spectral performance. In ethanol measurement, the relative delay was scanned from -300 fs to 300 fs within 300 ms, capturing almost all effective encoded information (AC components). A non-averaged photothermal interferogram is shown in Fig. 2(a), exhibiting a clean, low-noise baseline and clear broadband vibrational features. Performing an FFT on the interferogram in Fig. 2(a) yields the FT-OPT spectrum of ethanol, as shown in Fig. 2(c). The spectral coverage spans approximately 6000 – 9000 cm^{-1} , substantially exceeding the typical spectral bandwidth (~ 1000 cm^{-1}) accessible with conventional tunable sources. Pump-normalized spectrum shows excellent agreement with standard spectrum³⁰ (Fig. 2(c)). Ethanol exhibits three dominant C–H overtone peaks at ~ 8718 , ~ 8432 , and ~ 8292 cm^{-1} , along with a strong O–H overtone absorption band below ~ 7200 cm^{-1} . The C–H overtone features are accurately preserved even without normalization due to the flat pump spectral profile. Minor deviations below 7200 cm^{-1} may originate from pump spectral nonuniformity and polarization-dependent spectral response variations. Fig. 2(d) shows the FT-OPT interferogram of 1-octyne. The overtone vibration at ~ 6530 cm^{-1} exhibits a long lifetime, corresponding to a narrow spectral feature (FWHM ~ 29 cm^{-1}). FT-OPT is capable of recovering this lifetime-limited linewidth. In this case, the 4 ps temporal rectangular window corresponds to a spectral resolution of 10 cm^{-1} . This combination of spectral performance is difficult to achieve using frequency-domain excitation.

To evaluate the sensitivity, 1,7-octadiene was diluted in n-octane, and FT-OPT spectra (with a 4 ps temporal rectangular window) of solutions at different concentrations were measured in the low-concentration regime. The linear fit of the spectral peak intensity of 1,7-octadiene at ~ 6534 cm^{-1} versus concentration is shown in Fig. 2(e), with inset spectra at different concentrations. The nonzero signal at zero concentration (pure n-octane) is likely attributed to system background and the weak absorption baseline of n-octane. The standard deviation σ of the signal was calculated in a non-absorbing spectral region, and the system limit of detection (LOD) was estimated to be 3.4 mM (0.044% v/v) using the 3σ criterion³¹. By employing photothermal detection, FT-OPT effectively overcomes the challenge of weak overtone signals. This sensitivity is comparable to previously reported OPT and coherent Raman scattering (CRS) microscopy¹⁶, demonstrating the high sensitivity of the system. The current LOD is primarily limited by the noise of the probe and the pump power.

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FT-OPT further demonstrates strong molecular specificity across different chemical species. Pump-normalized spectra of alcohols and representative organic compounds reveal distinct broadband fingerprints, as shown in Fig. 2(f). Alcohols show characteristic O–H overtone absorption band with C–H features, while dimethyl sulfoxide (DMSO) and lipid-based samples exhibit distinct C–H signatures reflecting differences in carbon-chain structure. In Fig. 2(f), methanol, isopropanol (and also ethanol), DMSO, and oil exhibit closely spaced main peaks at ~8364, ~8432, ~8497, and ~8267 cm^{-1} , respectively, demonstrating high spectral discrimination capability. Weak shoulder peaks around 8500~8700 cm^{-1} from the three different oil samples are clearly resolved, rather than being masked or smoothed by the broad and strong main peak. This feature corresponds to the overtone vibration of unsaturated C–H bonds^{32,33}, a spectral detail that was not accessible in previous OPT microscopy¹⁷.

Capability of enhanced spatial resolution and volumetric imaging are further demonstrated. We evaluated the lateral resolution using a knife-edge imaging method at the interface between Olympus oil and glass (Fig. 2(g)), while simultaneously showing chemical contrast between materials with similar refractive indices. Assuming a Gaussian PSF, the FT-OPT signal profile along the dashed line was fitted with an error function to reconstruct the PSF as shown in Fig. 2(i). The lateral spatial resolution was determined to be ~885 nm from the FWHM of the PSF. The diffraction-limited spatial resolution determined by the pump, calculated using its average wavelength, is $0.61 \times 1350 \text{ nm} / 0.65 \approx 1267 \text{ nm}$. Therefore, the system is capable of imaging with a spatial resolution surpassing the pump-wavelength diffraction-limited optical resolution. Using the same method, volumetric imaging of the interface was performed to estimate the axial resolution, as shown in Figs. 3(h) and 3(j), yielding an axial spatial resolution of approximately 6.79 μm , which is consistent with the pump’s depth of focus.

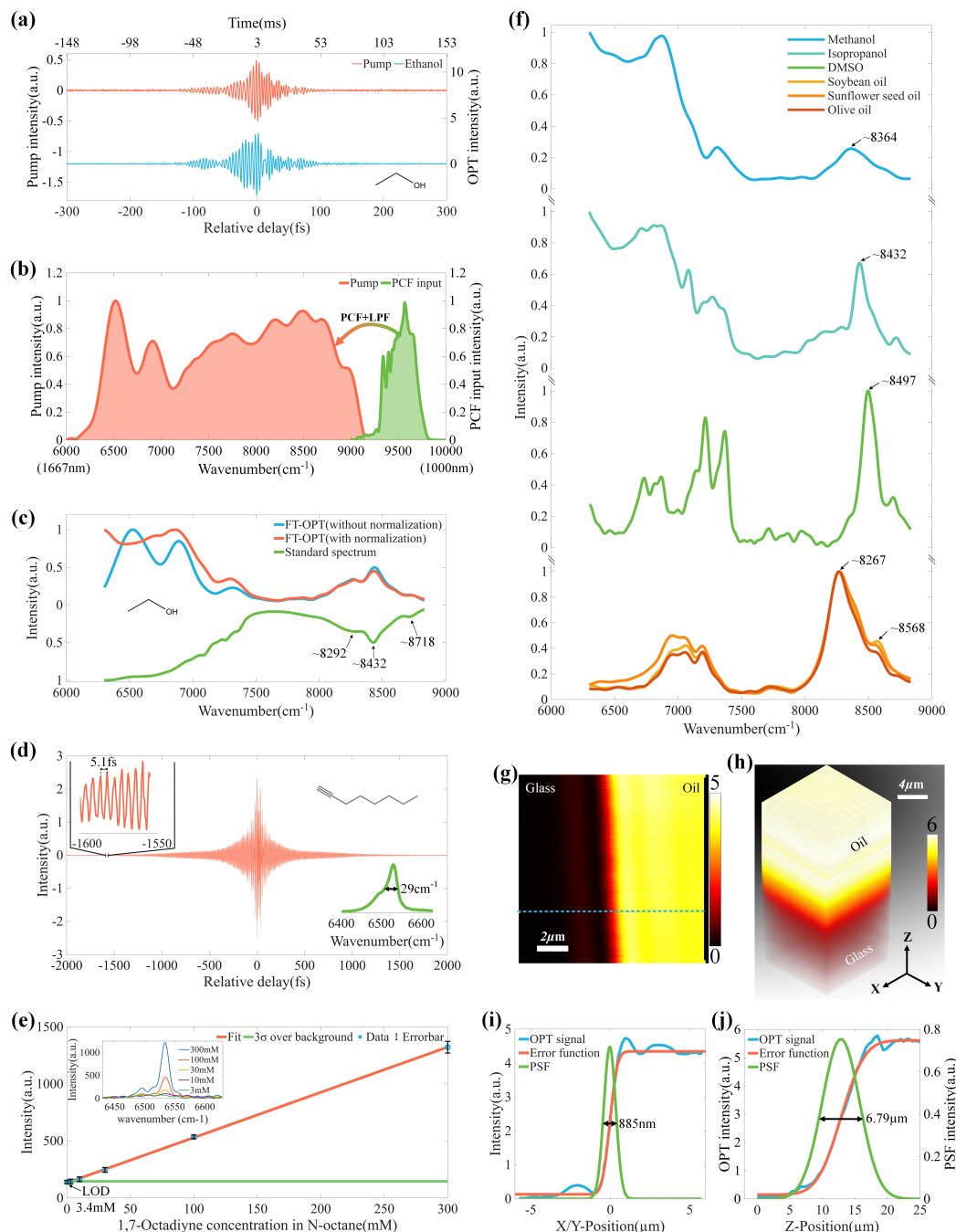


Fig. 2 Spectral and microscopic performance of FT-OPT. (a) First-order autocorrelation of the SWIR supercontinuum pump and FT-OPT interferogram of ethanol. (b) Pump spectrum retrieved from (a) by FFT, along with the PCF input spectrum. (c) FT-OPT spectrum of ethanol, compared with the reference spectrum. (d) FT-OPT interferogram of 1-octene; insets show a zoomed interferogram and the sharp peak at $\sim 6530\text{ cm}^{-1}$. (e) Linear fit of the FT-OPT signal versus 1,7-octadiene concentration in n-octane ($R^2 = 0.9998$) and system LOD evaluation. Each data point and its error bar represent the mean and standard deviation of 18 measurements, respectively. (f) pump-normalized FT-OPT spectra of alcohols and representative organic compounds. (g) FT-OPT XY-2D imaging of a knife edge (oil–glass interface). (h) FT-OPT XYZ-3D imaging of an oil–glass interface. (i, j) Error function fitting ($R^2 = 0.9968$ and 0.9976) for PSF extraction, yielding the system lateral and axial spatial resolutions. Except (d): 20 coherent averages; others: no averaging.

2.4 FT-OPT for molecular identification and unmixing

In addition to weak signal intensity, broad linewidths and severe spectral overlap in the overtone region are commonly considered to limit chemical specificity. Previous studies have used sophisticated algorithms to improve molecular discrimination³⁴, but concerns remain about their stability and generalizability. FT-OPT addresses this issue by using the supercontinuum spectral coverage to access multiple overtone vibrational modes, combined with high spectral resolution. To evaluate these advantages, C8 hydrocarbons were selected as a representative system. As key feedstocks and intermediates in fuels, organic materials, and biomolecules, hydrocarbons are of significant scientific interest. Fig. 3(a) shows the FT-OPT spectra of seven molecules (n-octane, isooctane, cyclooctane, 1-octene, 1,7-octadiene, 1-octyne, and ethylbenzene), which exhibit overall similar spectra but distinguishable local variations. The spectra are further divided into a low-wavenumber red region (6300~7750cm⁻¹) and a high-wavenumber yellow region (7750~9000cm⁻¹), corresponding to the spectral coverage of two common tunable SWIR systems (Er- and Yb-doped).

For each substance, 20 non-averaged FT-OPT spectra with a spectral resolution of 10 cm⁻¹ were acquired and subjected to principal component analysis (PCA). The 3D projection results are shown in Fig. 3(b–d), corresponding to the low-wavenumber region, high-wavenumber region, and full spectral range, respectively. The low-wavenumber region yields the poorest classification performance, with most species showing no clear separation in PCA space except isooctane and 1-octyne. The high-wavenumber region improves discrimination due to stronger spectral features, but partial overlap remains. In contrast, the full spectral range achieves clear clustering for all species, demonstrating the strong advantage of broadband coverage for molecular classification.

We further evaluated the spectral unmixing capability of FT-OPT. Three hydrocarbons were selected from the seven C8 species (including structurally similar alkane/alkene combinations) and mixed at predefined volume ratios. The FT-OPT spectra of the mixtures were then acquired and analyzed using the least absolute shrinkage and selection operator (LASSO) algorithm to determine the volume fractions of all seven candidate components. The corresponding unmixing absolute errors (at 10 cm⁻¹ spectral resolution) for the above three spectral ranges are shown in Fig. 3(e–h). The right panels illustrate the evolution of the error for each mixture as the spectral resolution degrades (10cm⁻¹ to 100cm⁻¹) with decreasing temporal rectangular window length. The low- and high-wavenumber regions exhibit complementary performance depending on the mixture

composition, whereas the full spectral range consistently achieves the lowest error across all mixtures. In most cases, the error increases significantly as the spectral resolution degrades. However, in a few cases such as Fig. 3(g), where the spectral differences among the mixture components (1,7-octadiene, 1-octyne, and ethylbenzene) are sufficiently large, higher spectral resolution is no longer necessary. Instead, shortening the temporal window to improve the signal-to-noise ratio (SNR) leads to more accurate unmixing results.

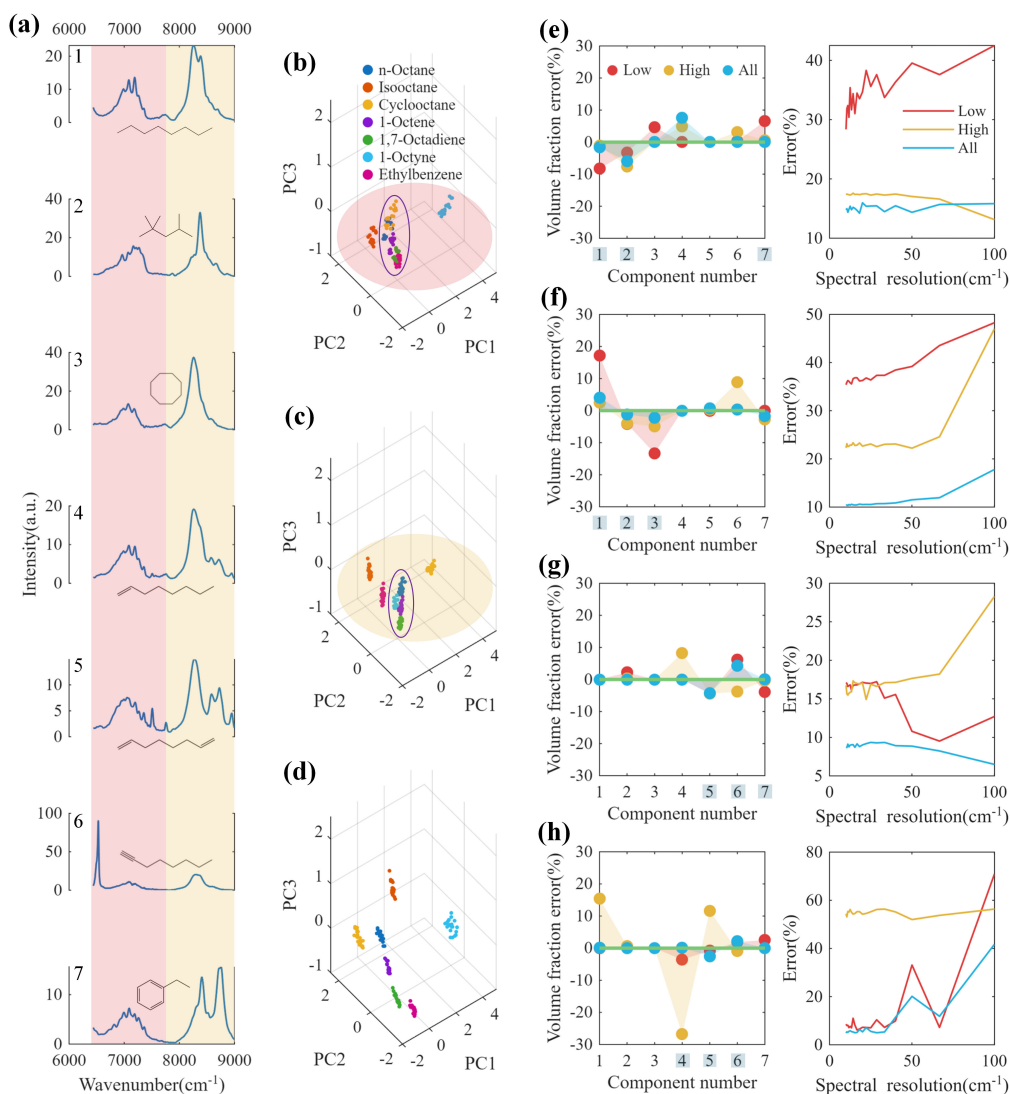


Fig. 3 Discrimination and unmixing of C8 hydrocarbons by FT-OPT. (a) FT-OPT spectra of several C8 hydrocarbons, where 1–7 correspond to n-octane, isooctane, cyclooctane, 1-octene, 1,7-octadiene, 1-octyne, and ethylbenzene. (b–d) PCA projections using spectral data from the red-shaded low-wavenumber region (b), yellow-shaded high-wavenumber region (c), and the full spectral range (d) in (a). (e–f) Unmixing errors of four mixture solutions using LASSO. Blue numbers indicate the dominant component in each mixture. Each mixture was unmixed using three spectral ranges in (a). Dependence of spectral unmixing error on degrading spectral resolution is shown on the right. FT-OPT spectral data were obtained with 20 coherent averages, except for PCA results (b–d) without averaging.

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285 *2.5 Microfluidic dynamic spectral monitoring of FT-OPT*

286 Although microfluidic photothermal detection for liquids or particles has been demonstrated³⁵, it
287 has so far been restricted to single-wavelength or spectrally discrete multi-wavelength operation.
288 To date, the integration with continuous broadband photothermal spectroscopy or overtone
289 spectroscopy remains unrealized. This is because the two spectroscopic techniques each impose
290 stringent requirements: rapid spectral acquisition or high sensitivity, respectively. FT-OPT enables
291 sensitive overtone spectroscopy within a few hundred milliseconds, thereby supporting dynamic
292 spectral monitoring. This provides a foundation for label-free spectral flow cytometry³⁶, which is
293 challenging to achieve with MIR approaches due to strong water absorption in aqueous
294 environments of cytometers. To verify this, we constructed a simple microfluidic system and used
295 a solenoid three-way valve to alternately switch water and cyclooctane through the focal volume
296 for dynamic FT-OPT tests. As shown in Fig. 4(a), water and cyclooctane form a distinct crescent-
297 shaped two-phase interface within the outer tubing and microchannel.
298 Intuitively, the flow velocity affects the modulation depth of the photothermal signal. An
299 excessively high velocity rapidly removes heat from the focal region, thereby limiting the
300 maximum temperature rise and accelerating thermal dissipation. To evaluate this effect, we
301 measured the photothermal responses under different flow velocities. Without relative delay
302 scanning, the APD signals were processed using match filtering³⁷ and normalized to a maximum
303 value of 3 V. The results are shown in Fig. 4(d,e), corresponding to water and cyclooctane,
304 respectively. This effect becomes pronounced when the velocity exceeds the order of hundreds of
305 millimeters per second. The thermal lens response exhibits a tendency toward saturation, with both
306 faster heating and accelerated thermal relaxation. Fig. 3(f) shows a decreasing photothermal
307 modulation depth for both substances with increasing flow velocity, further supporting this
308 conclusion. Since a flow velocity of 100 mm/s is already higher than that required for most
309 microfluidic applications, a velocity of 30 mm/s was adopted in the following experiments.
310 The spectral refresh period was set to ~0.842 s, yielding an effective duty cycle of ~51%, as the
311 relative delay scanning and stage turnaround each occupied nearly half of a cycle. The refresh rate
312 is currently limited by the stage reversal time and the optical chopper modulation frequency, the
313 latter constraining the maximum allowable relative delay scanning speed according to the Nyquist
314 sampling theorem. These limitations arise from the present hardware implementation rather than
315 the fundamental FT-OPT mechanism. Fig. 4(c) presents the spectral monitoring results acquired

over a period of ~ 10 s. Magnified views of two consecutive interferograms in Fig. 4 (c) are shown in Figs. 4(b). By segmenting the interferograms and performing FFT on each segment, dynamic FT-OPT spectra were obtained over a 59 s period, as shown in Fig. 4(g), comprising 35 spectra of water and 35 spectra of cyclooctane, and demonstrating excellent repeatability and stability. These results highlight the great potential of FT-OPT for high-throughput spectral identification.

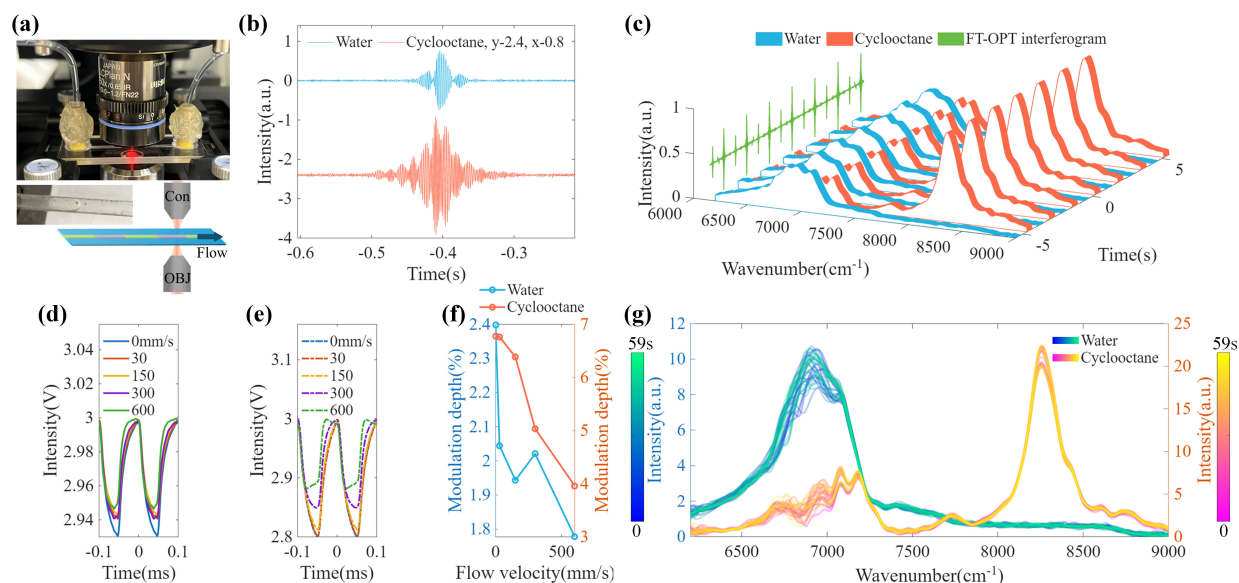


Fig. 4 Microfluidic molecular fingerprint monitoring using FT-OPT. (a) Experimental configuration and the oil–water interface within the microfluidic channel. (b) Magnified views of two FT-OPT interferograms in (c). The cyclooctane interferogram is offset along both the horizontal and vertical axes for clarity. (c) FT-OPT spectral monitoring over a 10 s period. (d, e) Photothermal modulation signals of water (d, solid lines) and cyclooctane (e, dashed lines) measured at different flow velocities. (f) Dependence of the photothermal modulation depth on flow velocity for water and cyclooctane. (g) Dynamic FT-OPT spectra acquired over 59 s, including 35 spectra of water and 35 spectra of cyclooctane. All spectra are plotted on the same axes; colorbar represents time. All data were obtained without averaging.

2.6 Label-free hyperspectral and volumetric imaging of FT-OPT

To the best of our knowledge, genuine time-domain excitation photothermal imaging has not been demonstrated. Existing approaches^{12,22} obtain high-resolution images by integrating all photothermal responses across the full broadband pump spectrum. FTIR is then applied to acquire spectra only at pixels of potential interest. Additionally, these techniques rely on thermosensitive fluorescent labels and cannot realize label-free detection. FT-OPT combines excellent spectroscopic capability with intrinsic photothermal imaging, offering distinct advantages for high-dimensional imaging in aqueous environments.

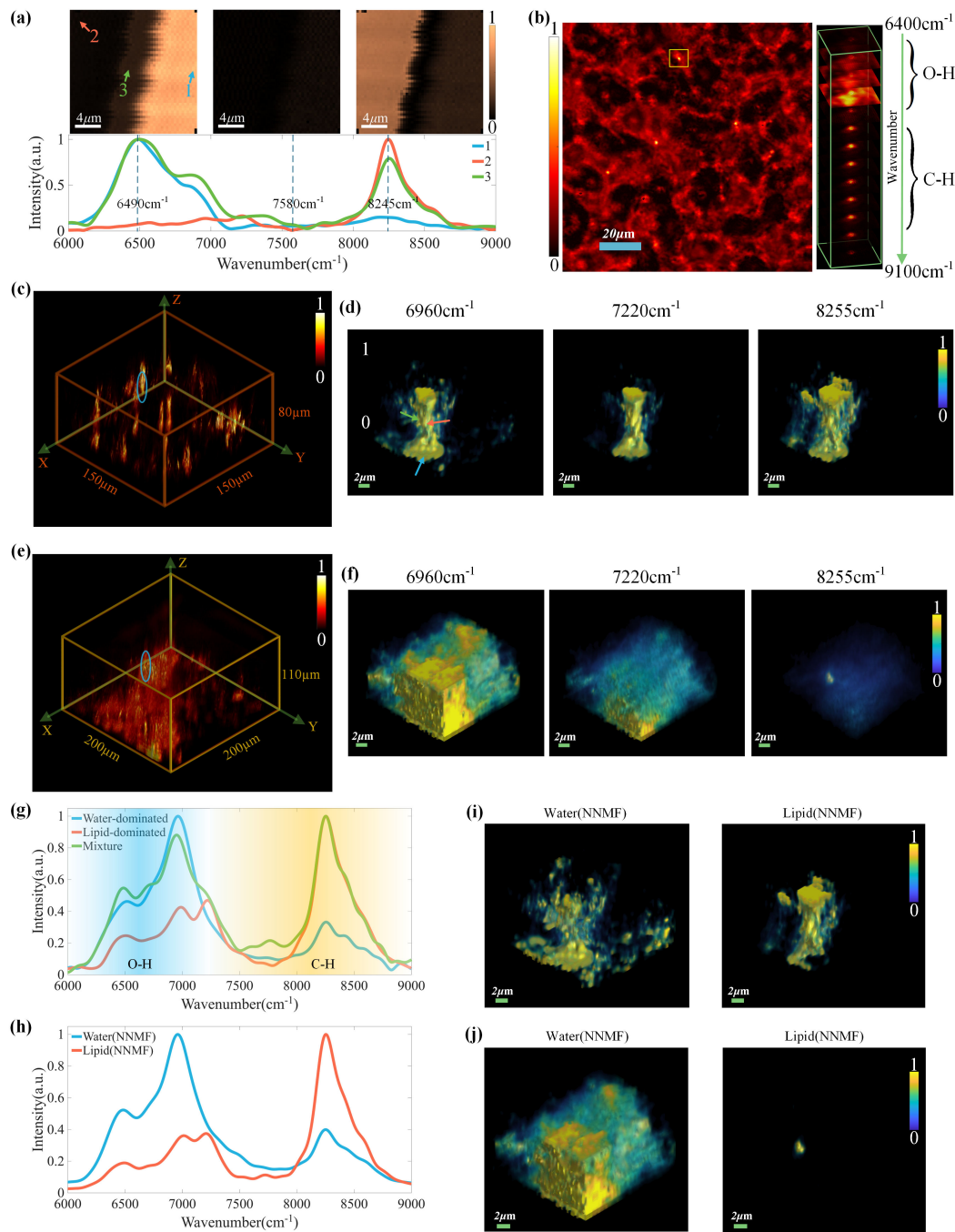


Fig. 5 Hyperspectral and Volumetric FT-OPT imaging. (a) FT-OPT hyperspectral imaging of the glycerol–olive oil interface. The blue, green and red arrows denote glycerol, olive oil and the transition region, respectively. (b) OPT image of a thin liver tissue and FT-OPT hyperspectral image of the region indicated. (c) OPT volumetric image of a thick liver tissue. (d) FT-OPT 4D hyperspectral image of the blue marked region in (c). The blue, red, and green arrows indicate regions dominated by water, lipids, and a mixture, respectively. (e) OPT volumetric image of a thick brain tissue. (f) FT-OPT 4D hyperspectral image of the blue marked region in (e). (g) FT-OPT spectra at arrows marked in (d). (h) Spectra of the two principal components obtained by applying NNMF to the 4D matrix in (d). (i) 3D concentration distributions of the liver tissue (d). (j) 3D concentration distributions of the brain tissue (f). For clarity, all 4D images are displayed with the Z-axis downscaled by a factor of 3. All data were obtained without averaging.

To validate the chemical selectivity of FT-OPT imaging, hyperspectral imaging was implemented at the glycerol–olive oil interface (Fig. 5a). The three panels from left to right correspond to 6490, 7580, and 8245 cm^{-1} , presenting images of the glycerol overtone peak (O-H), silent wavenumber, and olive oil overtone peak (C-H), respectively. The FT-OPT spectra at the three arrow-indicated positions are shown in the inset below. Distinct chemical contrast between them is clearly observed, with both overtone peaks simultaneously detected in the interfacial transition region (green arrow 3). We next extended the imaging to biological tissues immersed in a PBS buffer layer. Fig. 5(b) shows a non-spectrally resolved OPT image of a thin mouse liver tissue ($\sim 10\ \mu\text{m}$), which integrates the photothermal response of broadband pump. Tissue morphology and sparse micron-scale bright lipid droplets are observed. The right panel shows a hyperspectral image of the boxed region, revealing distinct spectral differences between the C–H and O–H overtone regions, corresponding to organic- and water-dominated distributions, respectively.

Fig. 5(c) displays a 3D view of a non-spectrally resolved OPT volumetric image of a thick liver tissue ($\sim 200\ \mu\text{m}$), clearly visualizing the distribution of lipid droplets. Although the lipid droplets appear axially elongated due to the mismatch between lateral and axial PSFs, the system achieves a penetration depth of $\sim 100\ \mu\text{m}$ in aqueous environments. In contrast, this remains significantly challenging for MIR methods⁵. In the current system, the penetration depth is limited by the high NA of the objective and the short probe wavelength. FT-OPT further enables 4D hyperspectral imaging. Fig. 5(d) presents 4D imaging results of the blue-marked region in Fig. 5(c), with volumetric slices displayed at three selected wavenumbers. Blue, red, and green arrows indicate water-dominated, lipid-dominated, and transition regions, respectively, and their FT-OPT spectra are shown in Fig. 5(g). The 3D and 4D imaging were also performed on a thick mouse brain tissue ($\sim 200\ \mu\text{m}$), and the corresponding results are shown in Figs. 5(e) and 5(f), respectively. The 3D image achieves a penetration depth comparable to that in the liver tissue. The 4D imaging was performed on the blue-marked region in Fig. 5(e), which corresponds to an organic-rich droplet embedded in an aqueous environment. Besides, probe transmission images reflecting light scattering of the above tissues are provided in Figs. S5–S7. Although the thin liver sample yields comparable morphological features via transmission, it cannot localize lipid droplet distributions. Meanwhile, the two thick tissues lack depth resolution owing to the single-photon interactions.

To further demonstrate the utility of hyperspectral data, we performed non-negative matrix factorization (NNMF) on the above 4D data cubes, assuming two principal spectral components:

water and organic constituents. Fig. 5(h) shows the spectra of the two components extracted from the liver tissue by the NNMF, while Fig. 5(i) displays the corresponding 3D distributions of their relative concentrations. Fig. 5(j) presents the algorithm results for the brain tissue, where the water background surrounding the droplet is effectively removed compared with Fig. 5(f).

3 Discussion and Conclusion

OPT or SWIR photothermal microscopy^{16,17} has recently been proposed to overcome the drawbacks of MIR photothermal approaches in aqueous environments and penetration depth, achieving remarkable success in biological and chemical imaging. Nevertheless, wavelength-scanned spectral acquisition limits its spectroscopic applications. Very recently, time-domain excitation has also been explored in MIR photothermal microscopy^{12,22}. However, its reliance on fluorescence labeling and extremely long integration times fails to exploit the advantages of time-domain excitation, while still inheriting the restrictions of MIR-based approaches. In this work, leveraging supercontinuum generation, we introduce FT-OPT to enable rapid, broadband, and high spectral resolution photothermal spectromicroscopy. FT-OPT also preserves the benefits of conventional OPT microscopy, including label-free detection, high sensitivity, submicrometer spatial resolution, volumetric imaging and compatibility with aqueous environments. Additionally, overtone spectroscopy has long been criticized for weak signals and poor specificity caused by spectral overlap. Here, we greatly enhance these signals by photothermal detection and obtain richer specificity information from the supercontinuum bandwidth—together lending greater confidence to overtone applications. We decouple spectral bandwidth from acquisition time, breaking a long-standing fundamental trade-off in photothermal field. Enabled by this advance, we demonstrate time-resolved microfluidic photothermal spectroscopy and label-free 4D photothermal imaging. These capabilities offer new tools for label-free molecular fingerprinting flow cytometry³⁶ and high-dimensional chemical imaging.

Although FT-OPT shows great application potential, considerable room remains for improving the current system. The current FT-OPT system has a spectral refresh time of ~ 0.842 s, which constrains its applicability to rapid hyperspectral imaging and dynamic microscopy. In fact, FT-OPT can provide substantially higher throughput for photothermal spectromicroscopy, and the present spectral throughput remains far from the limit set by the thermal response. According to the Nyquist sampling theorem, the relative delay scanning within one pump modulation period

must not exceed half the period of the highest-frequency component in the time-domain interferogram. Therefore, a higher modulation frequency permits faster relative delay scanning. In practice, the thermal response time of typical samples is on the microsecond scale³, indicating that the modulation frequency could reach the MHz range. The duty cycle of the relative delay scanning in the current system is ~51%, meaning nearly half the acquisition time yields no useful photothermal encoding. Employing high-speed galvanometers³⁸ or polygon scanners³⁹ would increase the delay scanning speed and improve the duty cycle toward ~100%, thereby enhancing acquisition efficiency. By replacing the LIA with advanced photothermal signal acquisition, such as single-pulse digitization⁴⁰, the estimated spectral refresh time could be reduced to ~464 μ s—nearly 2000 times faster than the present implementation. A detailed analysis of this projection is provided in the Supplementary Information.

Furthermore, in the current system, all spectral and spatial information is mapped exclusively onto the time axis, leading to an explosion in imaging time as dimensionality increases. It should be acknowledged that this is a common limitation of time-domain point-by-point scanning approaches for hyperspectral imaging. Nevertheless, wide-field approaches in MIR photothermal microscopy^{41–46} have effectively accelerated imaging, with beyond video-rate performance already demonstrated. A natural next step is to combine FT-OPT with these schemes and high-speed array detectors, so that a hyperspectral data cube could be acquired in a single relative delay scan. This could reduce the time for dynamic molecular fingerprint imaging to an unprecedented level.

Another key aspect of this work is the integration of SWIR supercontinuum generation with FT-OPT. This combination not only addresses the challenge of obtaining a broadband, high-brightness, and spectrally flat SWIR source, but also leverages the Fellgett advantage²³, thereby extending supercontinuum sources to photothermal field for the first time. Furthermore, due to the decoupling enabled by the time-domain excitation, the spectral coverage can be readily expanded without additional acquisition time. By increasing the pulse peak power coupled into the PCF or extending the PCF length^{28,29}, the pump spectral bandwidth could be further broadened, potentially covering the entire SWIR region and accessing a wider range of functional group vibrations^{20,24}. Combined with appropriate spectral unmixing or analysis algorithms^{17,39}, this would further enhance the molecular fingerprint interpretation capability of FT-OPT.

The sensitivity of FT-OPT can also be further improved. Using a probe source with lower shot noise, or exciting the first overtone regions¹⁶ can further increase the sensitivity by several fold.

Moreover, numerous sensitivity-enhancement strategies have been demonstrated, such as thermosensitive fluorescent labeling^{47,48}, nano-photothermal sensing^{49,50}, and metasurface-enhanced photothermal detection⁵¹. Additionally, contrast and spatial resolution can be enhanced by demodulating higher-order harmonics of the photothermal response^{37,52}. Regarding penetration depth, previous reports¹⁶ show that a probe beam with a lower NA and a longer wavelength can effectively suppress optical scattering in complex samples, enabling millimeter-scale penetration depths for OPT microscopy in biological tissues. Taken together, these compatible advancements position FT-OPT as a powerful label-free platform for high-throughput molecular fingerprinting dynamics, as well as complex biological and chemical imaging.

4 Materials and methods

4.1 SWIR supercontinuum generation

Pulses from a mode-locked ytterbium-doped fiber laser (100 MHz repetition rate, 1050 nm center wavelength, 95 fs pulse duration) were modulated at 10 kHz with a 50% duty cycle by a chopper (MC2000B, Thorlabs), expanded and shaped by a telescope, and then passed through an HWP–PBS–HWP configuration to control the power and polarization state. The beam (~175 mW after the chopper) was subsequently coupled into the PCF (SC-5.0-1040-PM, NKT Photonics, 1.93 m). The generated supercontinuum was collimated by an off-axis parabolic mirror (RFCAG-3.9-PC, Lbtek), and a long-pass filter (OFE1LP-1100, Jcoptix) was applied to isolate the SWIR portion of the spectrum as the pump source (for details, see Fig. S2).

4.2 FT-OPT spectroscopy and microscopy

The pump beam was combined with a CW beam using a DM (DMLP950, Thorlabs) and directed into a MI. A 1:1 BS (BSW29R, Thorlabs) divided the beam into a reference arm and a measurement arm. The reference arm remained stationary, while the measurement arm was mounted on a motorized scanning stage (UTS150PP, Newport) for relative delay scanning. At the MI output, one portion of the beam was sent to the microscopy module, and the other to PD1 (Model 1801, New Focus) and PD2 (PDA05CF2, Thorlabs), which provided the relative delay reference signal and the pump first-order autocorrelation, respectively. The raw signals from PD1

and PD2, along with the relative delay calibration, are provided in the Supplementary Information (Fig. S4).

Another CW beam was expanded by a telescope and served as the probe. The probe beam was combined with the MI output via a DM (DMLP650, Thorlabs). The relative defocus between the probe and pump beams at the sample was adjusted by varying the telescope lens separation to optimize the photothermal signal. The pump and probe beams were focused onto the sample using an objective (LCPLN50XIR, NA = 0.65, Olympus) to excite and detect the thermal lens effect. The transmitted probe beam was collected and collimated by a condenser (LCPLN50XIR, NA = 0.65, Olympus), passed through an adjustable aperture at the pupil plane, and then filtered by a short-pass and a band-pass filter (FESH0950 and FLH635-10, Thorlabs) to remove all wavelength components other than the probe. The resulting probe signal was detected using an APD (APD120A/M, Thorlabs). XY and Z scanning were performed using motorized translation stages (MLS203-1 and PLSZ, Thorlabs) at the sample plane and at the objective, respectively. The imaging parameters are listed in Tab. 1.

Tab. 1 FT-OPT imaging parameters

Imaging mode	Knife-edge	OPT	Hyperspectral	4D
XY pixel step	0.1 μm	0.4 μm	0.2 μm	0.4 μm
Z pixel step	0.45 μm	3 μm	\	3 μm
Pixel dwell time	1 ms	0.5 ms	842 ms	842 ms

4.3 Data acquisition and processing

The photothermal signal from the APD was demodulated using a LIA (LI 5640, NF) in external reference mode, with the reference signal provided by the chopper. The LIA output was filtered by an electronic low-pass filter (BLP-1.9+, Mini-circuits), displayed, and recorded by a 10-bit oscilloscope. The oscilloscope signal was then input into a computer program, where relative delay calibration and band-pass filtering were applied before FFT to obtain the spectrum. The raw APD and LIA outputs are shown in Fig. S4.

All FFTs were performed with a rectangular window (spectral resolution $\sim 1.2 / \text{windowing time}$). Zero-padding was applied on both sides of the time-domain interferograms up to ± 15 ps to obtain smooth spectral representations. With the exception of the C8 hydrocarbon analysis, windowing times as short as possible yet capable of capturing almost all time-domain encoded signals were

assigned to different samples: methanol, −210 to 210 fs; ethanol, −260 to 260 fs; isopropanol, −370 to 370 fs; DMSO, −500 to 500 fs; and three vegetable oils, −280 to 280 fs. For microfluidic spectral monitoring, a window of −400 to 400 fs was used, while for hyperspectral and 4D imaging, the window was set to −200 to 200 fs.

4.4 LOD measurement

The LOD is defined as the molar concentration at which the signal intensity reaches three times the standard deviation of the background. Assuming a linear relationship between signal and concentration, the LOD can be calculated as $3\sigma/S$, where σ represents the standard deviation of the background and S is the slope obtained from a linear fit of signal versus concentration. All spectra for LOD assessment in this work were acquired without any averaging.

4.5 Liquid samples

With the exception of microfluidic spectral monitoring, all liquid samples were confined to a ~120 μm layer using circular-aperture double-sided tape sandwiched between two standard coverslips. For knife-edge resolution testing, a sharp boundary was created by stacking two coverslips and securing them with transparent tape; the edge of the upper coverslip served as the sharp interface. To prepare the glycerol–olive oil interface, circular-aperture double-sided tape was attached to the first coverslip. Two small droplets of glycerol and olive oil were deposited on opposite sides of the circular hole, and a second coverslip was immediately overlaid. The sample was laid flat and left undisturbed for several hours until the interface stabilized. The volume mixing ratios of the four C8 hydrocarbon mixtures shown in Fig. 3 are listed in Tab. 2.

Tab. 2 Volume fractions of C8 hydrocarbon mixtures

Mixture (Fig. 3)	Volume fraction
Fig. 3(e)	n-octane, isooctane, ethylbenzene : [158, 645, 199]/1002
Fig. 3(f)	n-octane, isooctane, cyclooctane : [580, 160, 260]/1000
Fig. 3(g)	1,7-octadiene, 1-octyne, ethylbenzene : [330, 220, 455]/1005
Fig. 3(h)	1-octene, 1,7-octadiene, 1-octyne : [615, 290, 96]/1001

4.6 Microfluidic photothermal spectroscopy

The microfluidic chip was made of fused silica (75 mm × 24 mm × 3 mm) and contained a single channel 50 mm long, 1.5 mm wide, and 0.8 mm high. A 24 V three-way solenoid valve (LXRA2401000B, The Lee Company) was driven by a square wave to switch water and cyclooctane alternately into the channel in synchronization with the relative delay scan reversals. Fluid delivery and flow velocity were controlled by a syringe pump (LSP02-2B, Longerpump).

4.7 Mouse tissue samples

Mouse tissue sections were obtained from Wuhan Servicebio Technology Co., Ltd. For thin slices, the tissues were fixed in 4% PFA for at least 24 h. After fixation, the samples were dehydrated using a programmed sucrose series (total dehydration time: 12.5 h) as follows: 4 h in 4% PFA, 0.5 h in 15% sucrose solution, 1 h in 15% sucrose solution, 3 h in 30% sucrose solution, and 4 h in 40% sucrose solution. After dehydration, the tissue surfaces were briefly blotted to remove excess liquid, embedded in OCT compound, and rapidly frozen on a cryostat stage at -80 °C. Tissue sections of 10 μm thickness were cut using a cryostat, with each section placed on a separate microscope slide and stored at -80 °C until use. For thick slices, the processing steps were the same as for thin slices, with samples sectioned to 200 μm thickness using a vibratome. For imaging, tissue sections were thawed to room temperature and washed three times in PBS to remove residual PFA and OCT. The sections were then mounted using double-sided adhesive tape with circular openings and covered with a coverslip, resulting in tissue sections confined within a PBS layer between the slide and coverslip.

Disclosures

The authors declare no conflicts of interest.

Author contributions

G.L. conceived the idea, built the system, performed experiments and data analysis, and drafted the manuscript. R.H. and J.J. conducted the literature review and assisted with the experiments.

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M.L. proposed using PCF for spectral broadening in a previous study and offered suggestions for the system development. X.Lyu., R.A., and X.Lin. assisted with figure preparation and manuscript revision. T.L. provided suggestions on photothermal detection. Y.L. provided technical support. H.W. refined the concept, supervised and guided all aspects of the study.

Data availability

The data and code used in this research are available from the corresponding author upon reasonable request.

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**Supplementary Information for
Supercontinuum Fourier-transform Overtone Photothermal
Spectroscopy and Microscopy**

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This file contains the following contents:

- Figures S1-S7
- Fellgett advantage of FT-OPT
- Comparison of signal intensity between photothermal and absorption spectroscopy
- PCF coupling setup
- Broadband pump power calculation
- Raw signal and relative delay calibration
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- Probe transmission and OPT imaging of the thick liver tissue
- Probe transmission and OPT imaging of the thick brain tissue
- Estimated optimization of spectral acquisition speed

22 Fellgett advantage of FT-OPT

23 The Fellgett advantage¹, also known as the multiplex advantage refers to the improvement in SNR
 24 that arises when multiple spectral elements are measured simultaneously rather than sequentially.
 25 In FT-OPT, all vibrational states are encoded and detected by thermal lens sensing at once;
 26 therefore, it inherently benefits from the Fellgett advantage.
 27 Formally, under detector-noise-limited conditions, the SNR improves proportionally to the square
 28 root of the number of spectral elements N measured:

$$SNR \propto \sqrt{N} \quad (1)$$

30 For the system demonstrated in this work, the total pump power at the sample within the 6000–
 31 9000 cm^{-1} range is ~21 mW. For the LOD measurement of 1,7-octadiene in n-octane, a relative
 32 delay window of –2000 to +2000 fs was applied, corresponding to a spectral unit width of ~8.33
 33 cm^{-1} in the frequency domain (360 spectral units over the ~3000 cm^{-1} range). Accordingly, the
 34 average pump power allocated to each spectral unit is $21\text{mW}/360 \approx 58.3\mu\text{W}$. Under a frequency-
 35 domain excitation scheme—employing the same supercontinuum source with a tunable
 36 narrowband filter—the pump power of $58.3\mu\text{W}$ is inadequate to produce an appreciable
 37 photothermal signal, owing to the overtone transitions being several orders of magnitude weaker
 38 than the fundamental transitions.

39 It is worth noting that Eq. (1) is derived under detector-noise-limited conditions. In the present
 40 system, the shot noise of the probe beam (CW) has not been rigorously optimized or controlled;
 41 therefore, the SNR, or LOD, still has room for further improvement.

Comparison of signal between photothermal and absorption spectroscopy

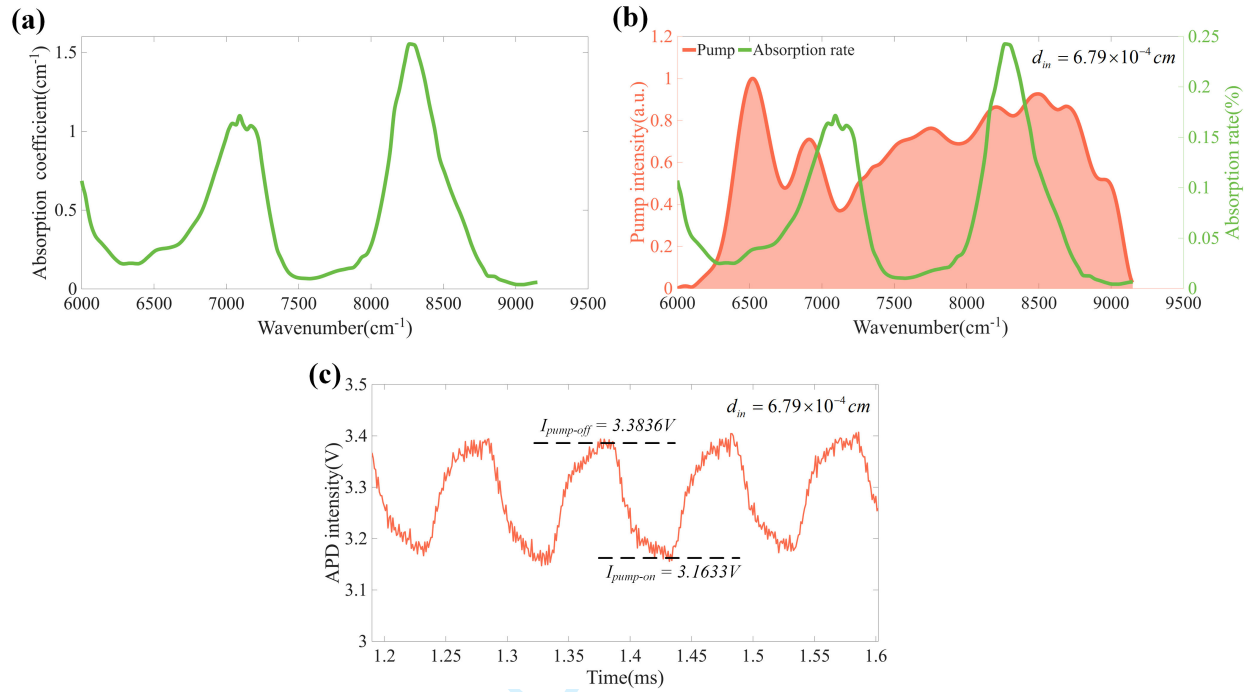


Fig. S1 (a) Absorption coefficient $\alpha(\omega)$ of lipids². (b) Pump power spectral density and lipid absorption percentage. (c) Raw photothermal signal of olive oil measured by the APD.

In addition to enabling optical sectioning capability, another major advantage of photothermal spectroscopy is its high sensitivity, particularly in spectromicroscopy applications. In these contexts, the interaction length between light and matter is extremely short (on the micrometer scale). Moreover, overtone transitions are several orders of magnitude weaker than fundamental transitions. According to the Beer–Lambert law, the fraction of incident light absorbed by the sample is therefore very limited, resulting in insufficient sensitivity for conventional spectroscopic methods. Let the absorption coefficient be $\alpha(\omega)$, the interaction length be d_{in} , and the power spectral density of the pump light be $S(\omega)$. Then, the fraction of the total energy remaining after the pump light interacts with the material is:

$$T = (\int 10^{-\alpha(\omega)d_{in}} S(\omega) d\omega) / (\int S(\omega) d\omega) \quad (2)$$

Taking lipid samples as an example, the absorption coefficient $\alpha(\omega)$ in the SWIR band is shown in Fig. S1(a), while $S(\omega)$ is provided in Fig. S1(b). The d_{in} is set to $6.79 \times 10^{-4} \text{ cm}$ (corresponding to the axial resolution of the system). The resulting signal intensity (modulation depth) is given by:

$$I_{M1} = 1 - T \approx 0.075\% \quad (3)$$

The raw photothermal signal of the lipid sample (olive oil) measured by FT-OPT is shown in Fig. S1(c), and its signal intensity (modulation depth) is:

$$I_{M2} = |I_{\text{pump-off}} - I_{\text{pump-on}}| / I_{\text{pump-off}} \approx 6.5\% \quad (4)$$

Therefore, the photothermal detection in this system enhances the signal modulation depth by approximately 86-fold, approaching two orders of magnitude. This enhancement can be further increased by raising the pump power.

Furthermore, conventional spectroscopic methods directly measure relative pump attenuation; hence, absorbance recovery is highly sensitive to fluctuations in pump spectral power. In contrast, photothermal approach relies on detecting variations in the probe beam, with the pump acting solely as an excitation source, thereby relaxing the stability requirements for the pump.

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PCF coupling setup

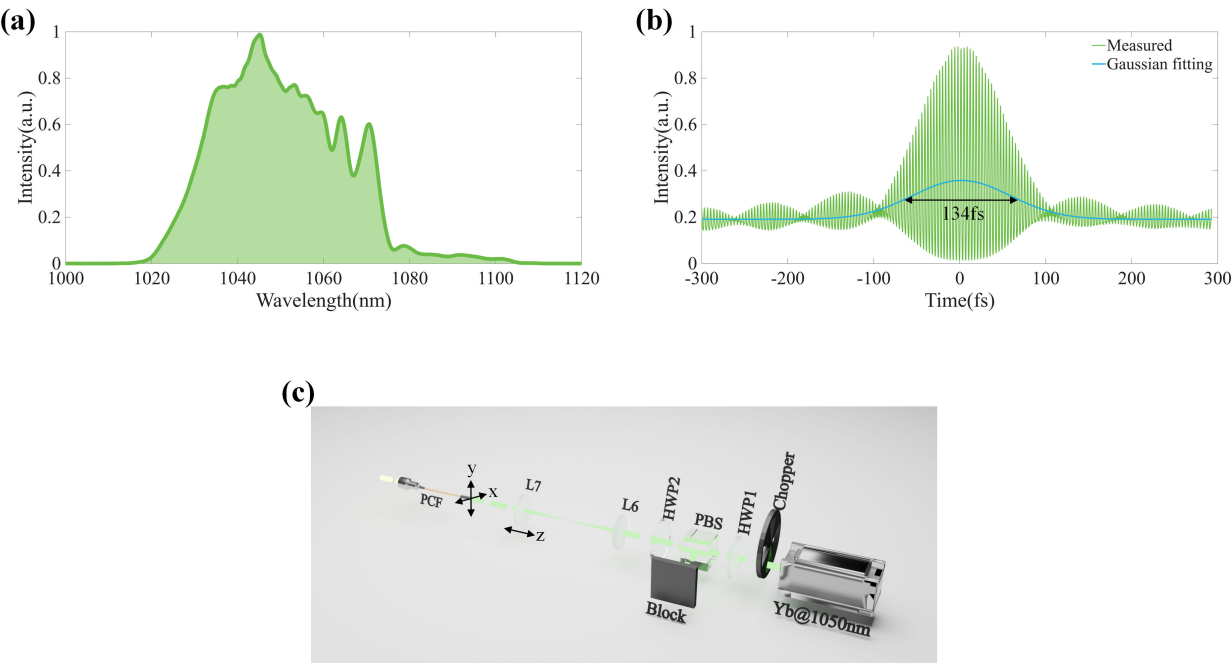


Fig. S2 Temporal and spectral characteristics of the pulses injected into the PCF and the coupling setup. (a) Spectrum of the injected pulses. (b) Intensity autocorrelation of the injected pulses. (c) PCF coupling setup.

The PCF was coupled using a gradient-index (GRIN) lens integrated in a single-mode fiber collimator, and a telescope system was employed to shape the beam. By carefully adjusting the lateral (XY) position of the GRIN lens and the axial (Z) position of the telescope system lens, optimal coupling efficiency can be achieved. HWP2 was adjusted to set the polarization state of pulses launched into the PCF, maximizing spectral broadening.

79 Broadband pump power calculation

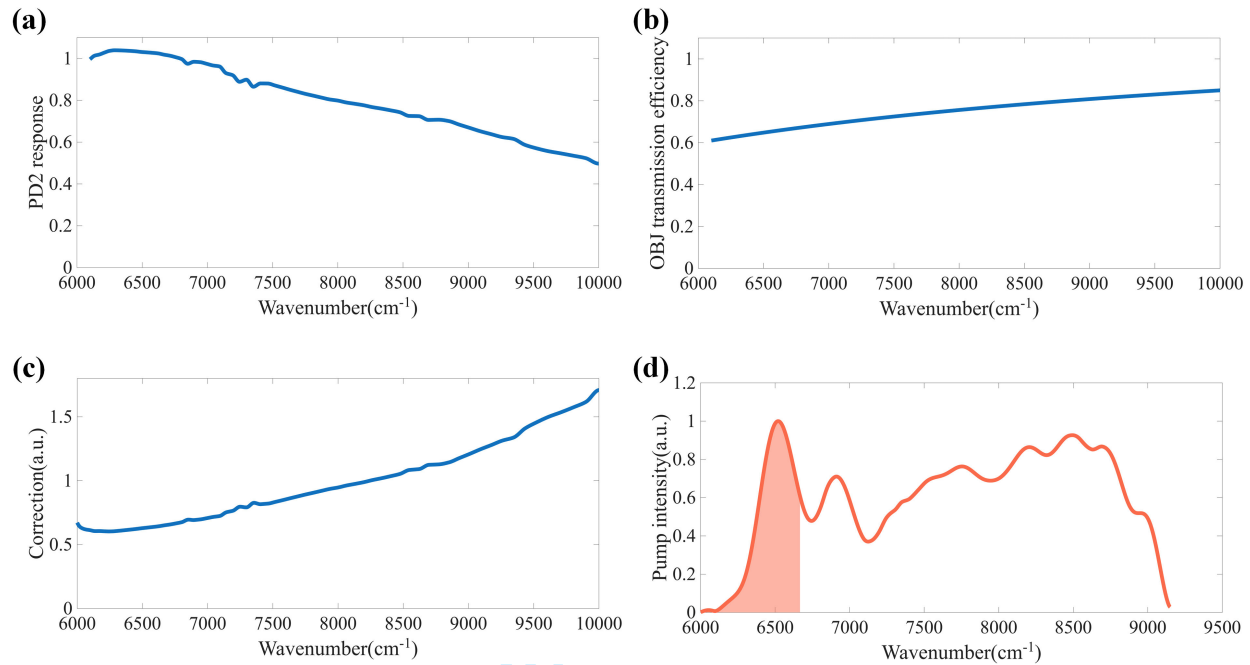


Fig. S3 Pump spectrum correction and power calculation. (a) Response curve of PD2. (b) Transmission curve of the objective. (c) Corrected pump spectrum at the sample, obtained by dividing the objective transmission curve by the PD2 response curve. (d) Calculation of the pump power, with the red-shaded area indicating the transmitted portion using a 1500 nm long-pass filter.

A standard optical power meter cannot directly measure broadband power. Before the objective, a 1500 nm long-pass filter (FELH1500, Thorlabs) was used to cut off the pump, and the transmitted portion was measured to be 5 mW. The objective transmission near 1500 nm is approximately 0.62, so the corresponding power at the sample is $0.62 \times 5 \text{ mW} = 3.1 \text{ mW}$, corresponding to the red-shaded area in Fig. S3(d). The ratio of the integral of the entire curve in Fig. S3(d) to the red-shaded portion is 6.84; therefore, the total pump power at the sample is approximately $6.84 \times 3.1 \text{ mW} = 21.204 \text{ mW}$.

91 Raw signal and relative delay calibration

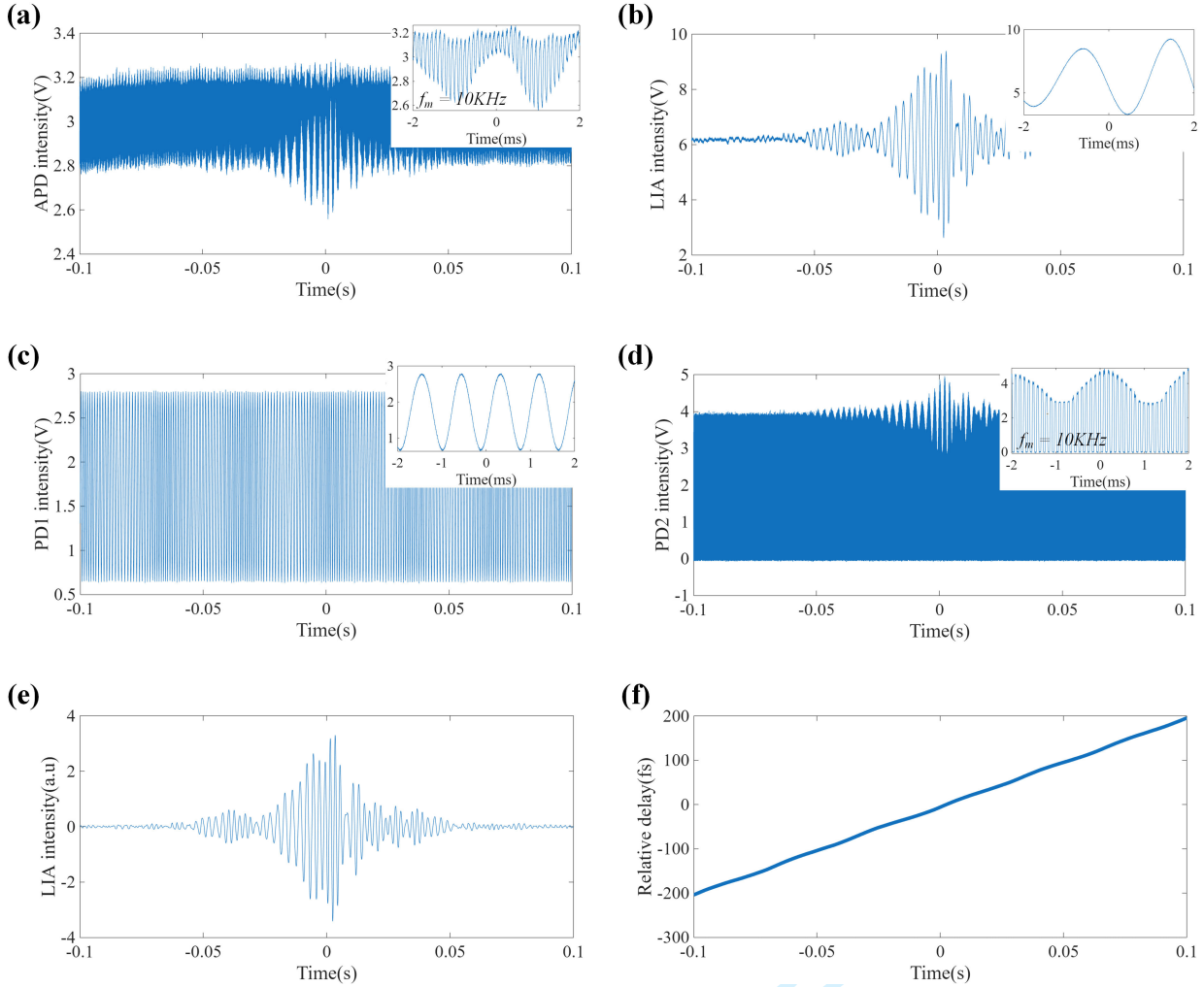


Fig. S4 Original signal and relative delay calibration (ethanol sample). (a) Raw signal from the APD. (b) Raw output from the LIA, demodulated from (a) using the external reference mode. (c) Raw signal from PD1, corresponding to the interference of the CW. (d) Raw signal from PD2. (e) Band-pass filtered version of the signal in (b). (f) Relative delay signal obtained by applying the Hilbert transform to (c) and performing phase unwrapping.

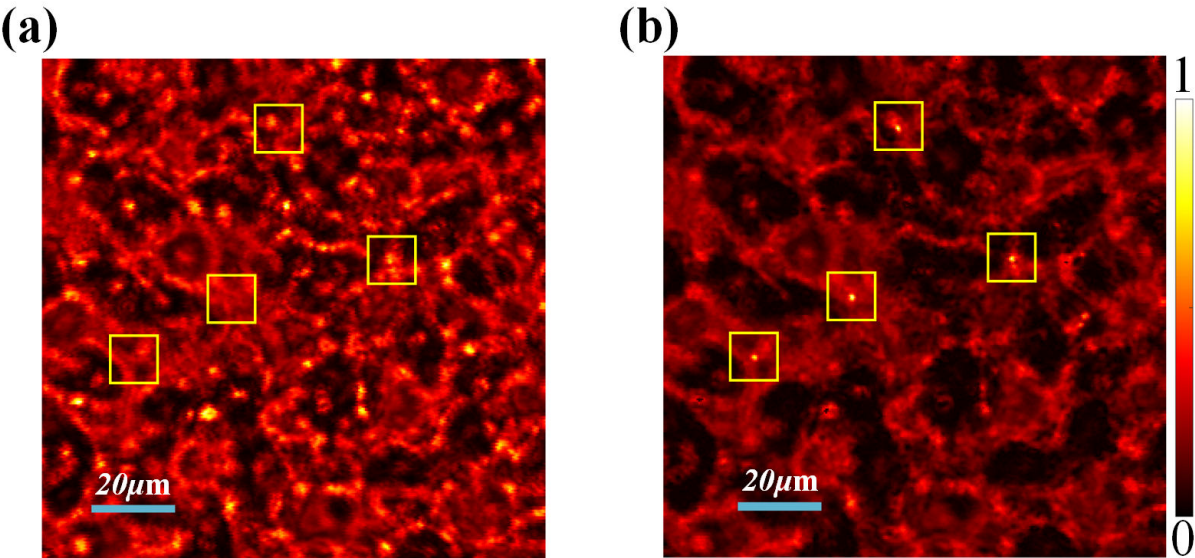
Due to the non-uniform motion of the motorized stage, jitter in the relative delay scanning can affect the accuracy of spectral reconstruction. By applying a Hilbert transform to the interference signal from PD1, the temporal variation of its phase can be extracted ($\varphi(t)$), allowing this jitter to be monitored and compensated. The resulting relative delay $\tau(t)$ is expressed in Eq. (5).

$$\tau(t) = \frac{\varphi(t)}{2\pi\nu_{cw}} + \tau_0 \quad (5)$$

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3 102 Here, ν_{CW} represents the frequency of the CW, and τ_0 denotes the offset, with the zero-delay point
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5 103 chosen at the maximum of the photothermal signal. After obtaining $\tau(t)$, a linear relative delay grid
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7 104 is reconstructed, and the time-domain signals are resampled onto this grid using interpolation.
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105 **Probe transmission and OPT imaging of the thin liver tissue**



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107 **Fig. S5 |** Comparison between probe transmission imaging (a) and photothermal imaging (b) of the thin liver tissue. The two exhibit
108 similar histomorphological features. Yellow boxes denote regions with sparse distribution of lipid droplets, which cannot be
109 well-resolved in transmission imaging (a).

110 Fig. S5(b) (same as Fig. 5(b)) shows the OPT image of the thin liver tissue, while Fig. S5(a) is the
111 corresponding probe transmission image. The APD and photothermal signals are acquired
112 synchronously pixel by pixel. Low-pass filtering the APD signal removes the chopper frequency;
113 the resulting quasi-DC signal forms the probe transmission image. Owing to the thin sample (single
114 or few cell layers), the transmission image provides refractive index/scattering contrast. The
115 morphologies in Figs. S5(a) and S5(b) agree well, with the photothermal image showing a sparse
116 distribution of lipid droplets.

117 **Probe transmission and OPT imaging of the thick liver tissue**

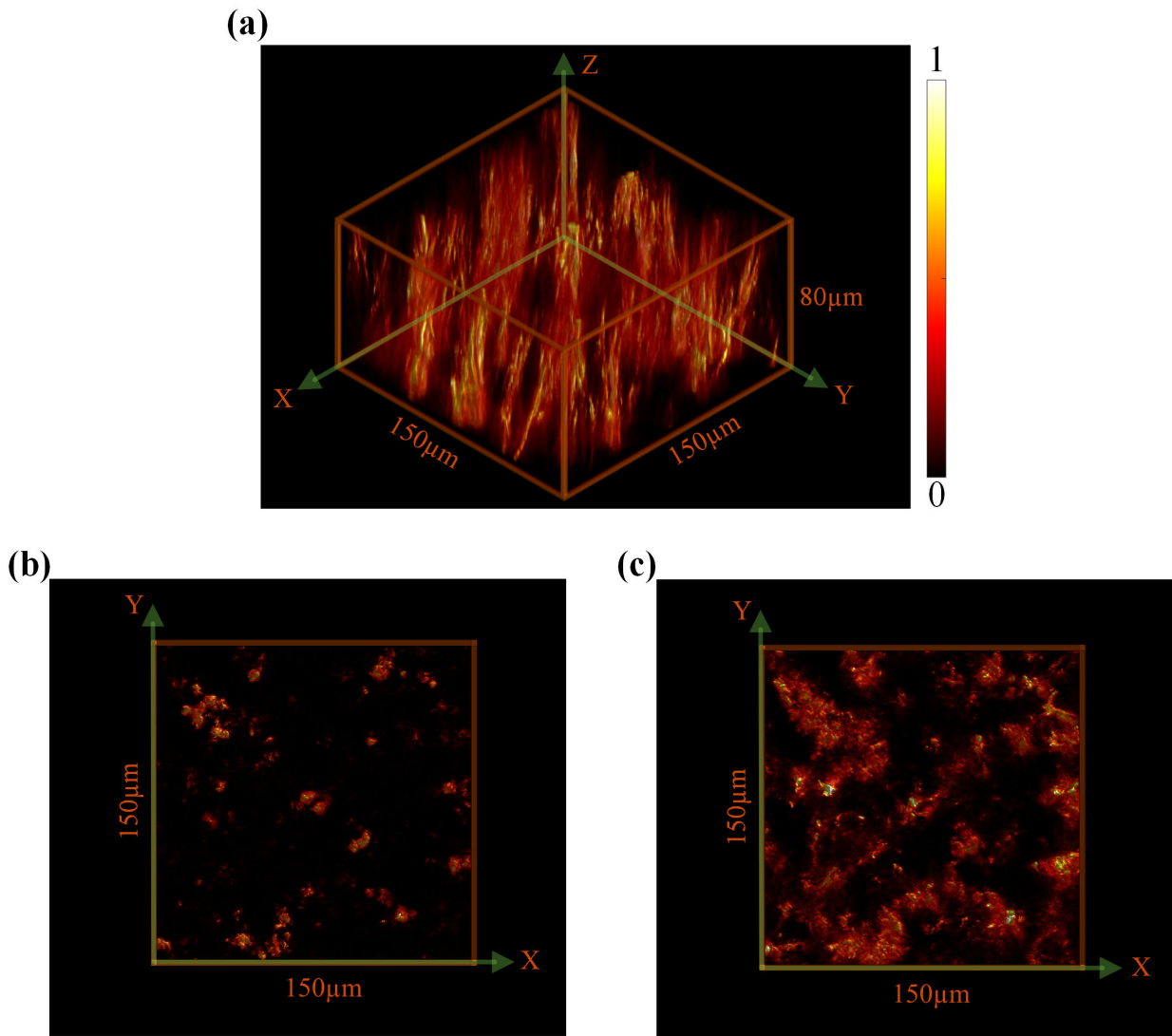


Fig. S6 Comparison between probe transmission imaging and photothermal imaging of the thick liver tissue. (a) Probe transmission volumetric imaging of a thick liver tissue sample, where axial resolution cannot be achieved. (b) and (c) Top-view images of photothermal and probe transmission volumetric imaging of the same region, respectively.

Probe transmission and OPT imaging of the thick brain tissue

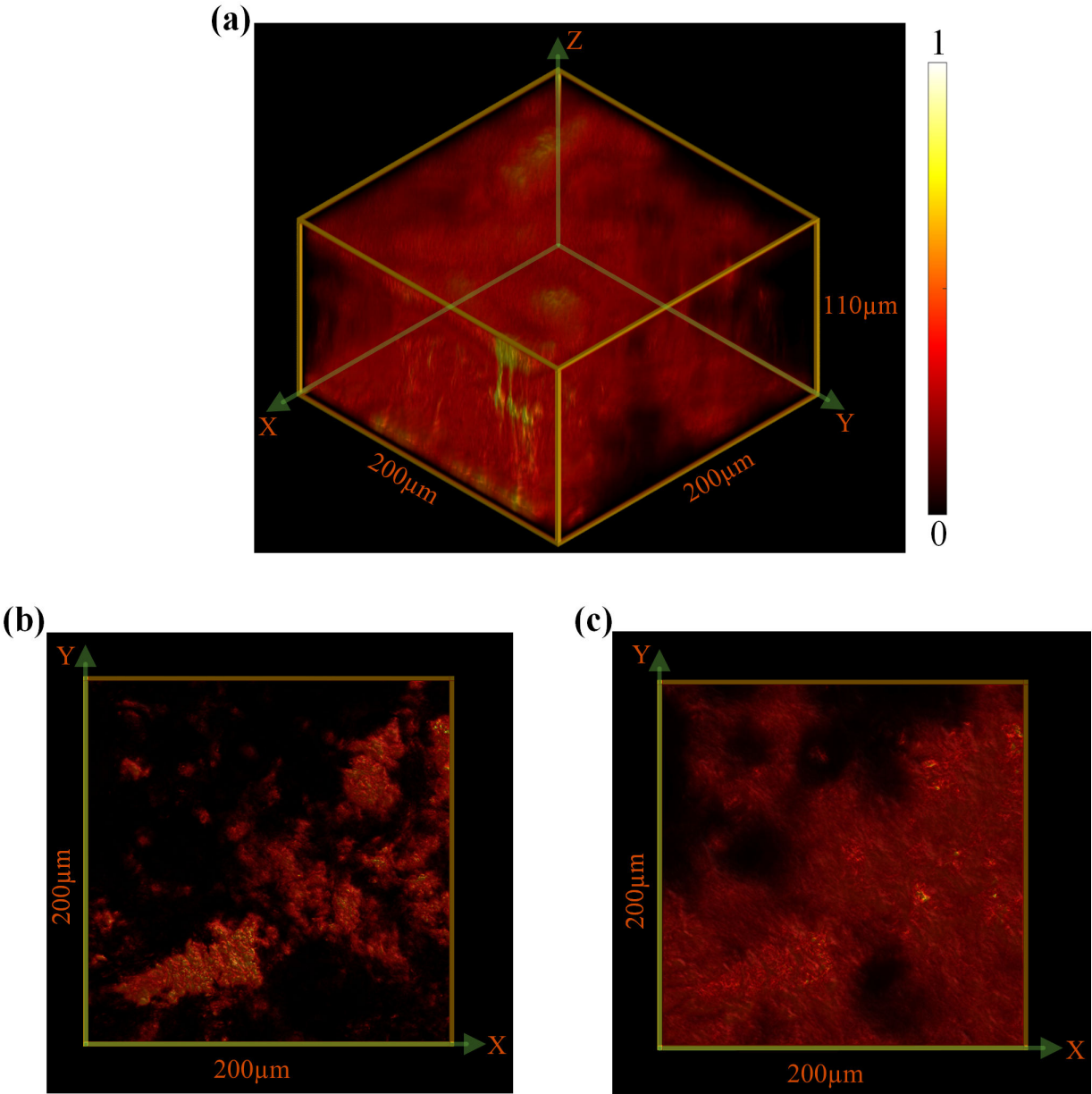


Fig. S7 Comparison between probe transmission imaging and photothermal imaging of the thick brain tissue. (a) Probe transmission volumetric imaging of a thick brain tissue sample, where axial resolution cannot be achieved. (b) and (c) Top-view images of photothermal and probe transmission volumetric imaging of the same region, respectively.

127 Estimated optimization of spectral acquisition speed

128 Here, we assume that the relative delay changes linearly over time. Currently, the spectral
129 acquisition rate is limited by the chopper modulation frequency ($f_m = 10 \text{ KHz}$). By the Nyquist
130 sampling theorem, the maximum frequency $f_{\max 1}$ of the FT-OPT time-domain interferogram is
131 given by:

$$f_{\max 1} = \frac{f_m}{2} = 5000 \text{ Hz} \quad (6)$$

133 Assuming that the highest wavenumber to be detected is 9000 cm^{-1} , the corresponding maximum
134 allowable relative delay scanning speed is:

$$V_{\tau-\max 1} = \frac{f_{\max 1}}{9000 \text{ cm}^{-1} \times 100 c_0} \approx 18519 \text{ fs / s} \quad (7)$$

136 Here, c_0 denotes the speed of light in vacuum. Thus, increasing the modulation frequency f_m can
137 effectively accelerate the relative delay scanning speed. However, the above inference assumes
138 ideal conditions. In practical applications, the demodulated signal from the LIA is typically used.
139 While this effectively suppresses noise, it further constrains the maximum achievable relative
140 delay scanning speed. The current LIA time constant is set to $\tau_{LIA} = 0.1 \text{ ms}$ (one period of the
141 modulation frequency f_m), corresponding to an upper frequency limit $f_{\max 2}$ of the signal to be
142 demodulated and a maximum relative delay scanning speed given by:

$$f_{\max 2} = \frac{1}{2\pi\tau_{LIA}} \approx 1591 \text{ Hz} \quad (8)$$

$$V_{\tau-\max 2} = \frac{f_{\max 2}}{9000 \text{ cm}^{-1} \times 100 c_0} \approx 5892 \text{ fs / s}$$

144 In the FT-OPT system presented in this work, the relative delay scanning speed is set to 2000 fs/s ,
145 providing a conservative margin ($\eta = 5892/2000 = 2.946$) to avoid potential spectral leakage
146 caused by noise and the nonlinearity of the relative delay scanning.

147 In summary, there are two approaches to further increase the spectral acquisition speed. The first
148 is to increase the pump modulation frequency and the duty cycle of the relative delay scanning.
149 Assuming that an AOM or EOM is used to modulate the pump at $f_{ml} = 1 \text{ MHz}$ —approaching the
150 upper limit of the photothermal response speed of typical samples³—and a high duty-cycle
151 ($D_I \sim 100\%$) relative delay scanning scheme (e.g., galvanometric mirrors⁴ or polygon scanners⁵) is
152 employed, the resulting acceleration factor β_I in spectral acquisition rate, compared to the current
153 system configuration ($f_m = 10 \text{ KHz}$, $D \sim 51\%$), can be expressed as:

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$$\beta_1 = \frac{f_{m1} D_1}{f_m D} \approx 196 \quad (9)$$

155 The second is using a more advanced demodulation technique instead of the LIA. In the field of
156 MIR photothermal microscopy, fast digitization has already been used to detect the transient
157 photothermal signal induced by a single modulated pulse⁶. When applied to FT-OPT, this
158 technique allows the maximum frequency of the time-domain interferogram to revert to the
159 expression in Eq. (6), yielding an acceleration factor β_2 given by:

160
$$\beta_2 = \frac{2\pi\tau_{LIA}f_m}{2} \approx 3.14(\tau_{LIA}f_m = 1) \quad (10)$$

161 As a result, under ideal conditions, the overall acceleration factor is:

162
$$\beta = \eta\beta_1\beta_2 \approx 1813 \quad (11)$$

163 Compared with the current single spectrum refresh time of ~0.842 s, the accelerated value is
164 reduced to ~464 μ s.

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